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Investigation Filters

Risk Category

High ×Critical ×ⓧ⌵

Recommended Action

MANUAL REVIEW ×BLOCK ×ⓧ⌵

Abuse Sentinel

AI-Powered Fraud Detection using Machine Learning + Graph Analytics + Risk Propagation + Explainable AI

Transactions

Fraud Cases

Critical

High Risk

Blocked

50,0001,2131,0563141,056

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Fraud Investigation Queue

	transaction_id	user_id	amount	ml_fraud_probability	graph_risk_normalized	bfs_propagated_risk	unified_risk_score	risk_category	recom
9309	T032055	U02920	1040.09	0.9924	0.625	1	90.1701	Critical	BLOCK
27437	T008098	U00572	1092.92	0.9924	0.625	1	90.1681	Critical	BLOCK
24740	T039591	U03022	1936.97	0.9891	0.625	1	89.9726	Critical	BLOCK
2163	T024021	U02120	13755.18	0.9853	0.625	1	89.7424	Critical	BLOCK
2166	T023687	U02120	24327.28	0.9823	0.625	1	89.5637	Critical	BLOCK
38956	T043249	U04163	260.08	0.9996	0.5833	1	89.5622	Critical	BLOCK
37131	T046621	U04163	125.03	0.9991	0.5833	1	89.5271	Critical	BLOCK
24198	T020389	U00861	1221.19	0.9987	0.5833	1	89.5037	Critical	BLOCK
31179	T035127	U04163	145.35	0.9986	0.5833	1	89.501	Critical	BLOCK
26541	T037452	U04163	221.17	0.9986	0.5833	1	89.4976	Critical	BLOCK

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Investigation Filters

Risk Category

High × Critical × ⌵

Recommended Action

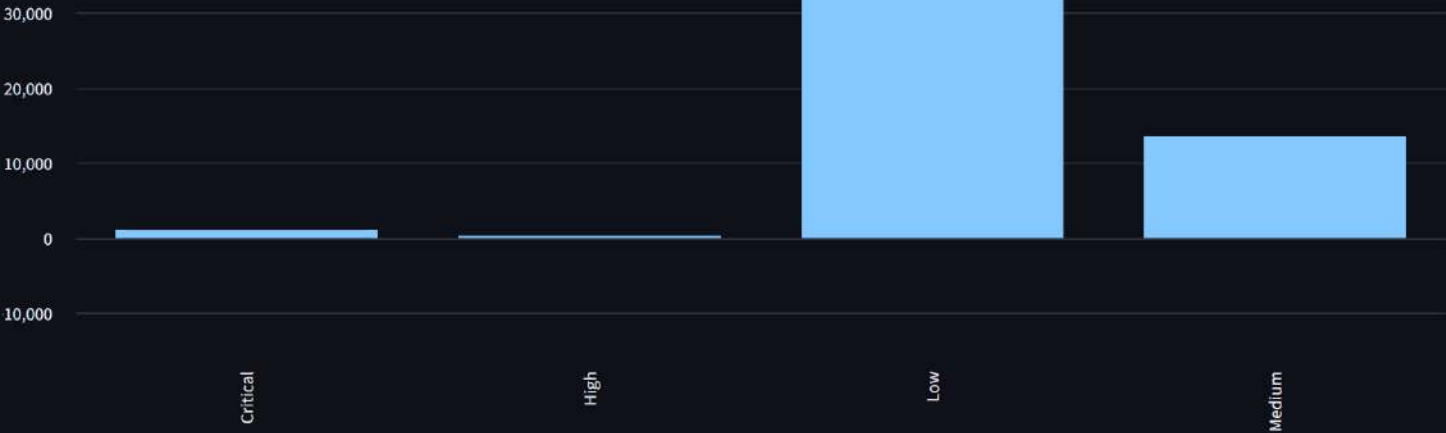
MANUAL REVIEW × ⌵

BLOCK ×

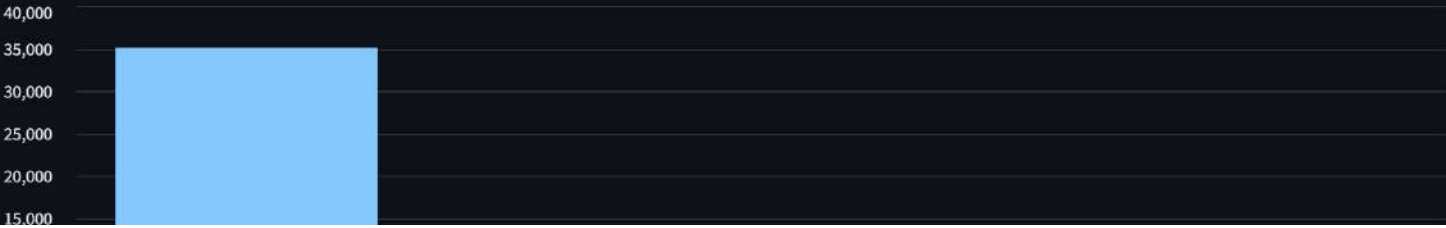
36905T027616U02433279.130.98890.5833188.9158CriticalBLOC

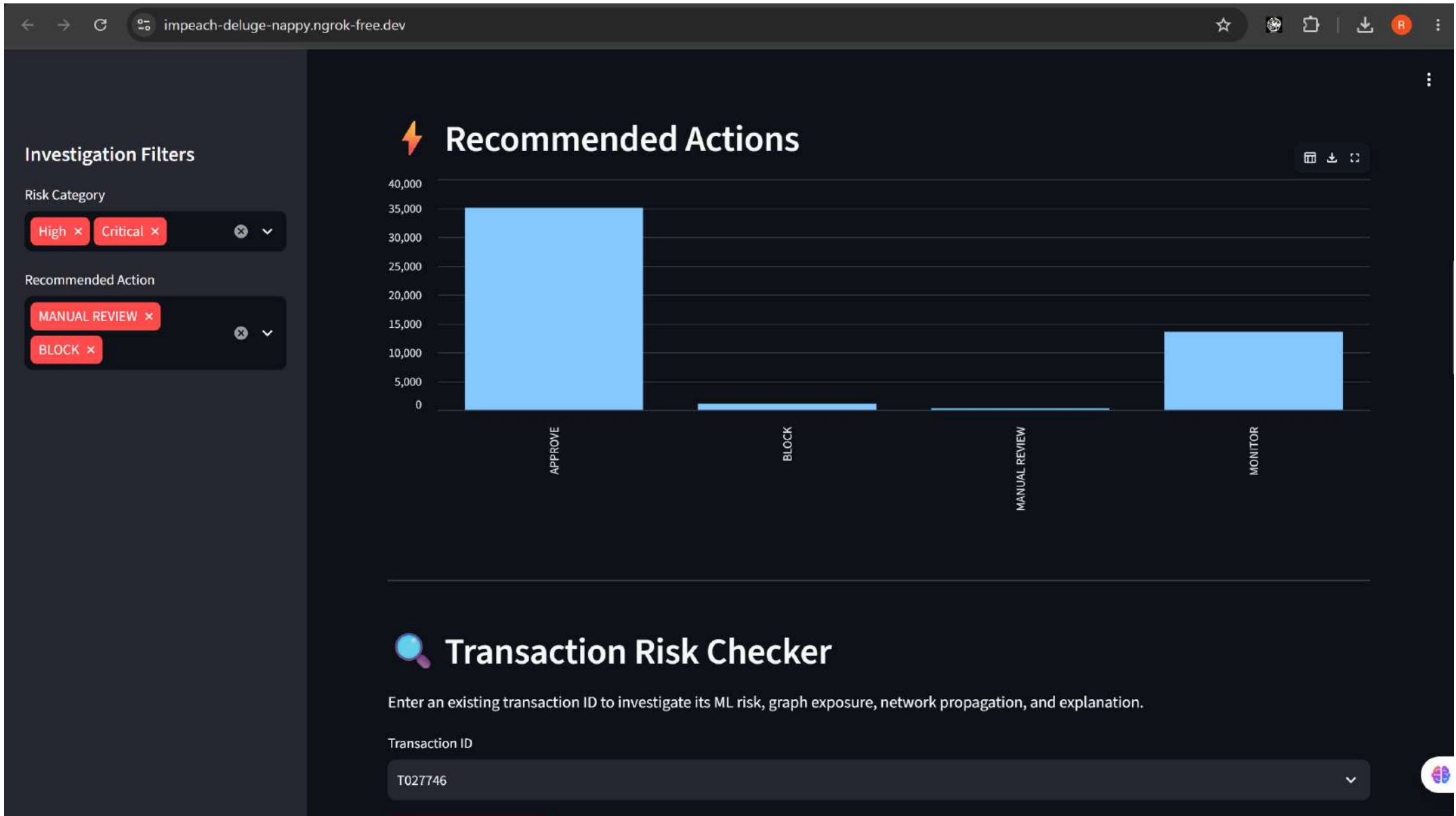
 Risk Distribution





 Recommended Actions





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Investigation Filters

Risk Category

High × Critical × × ▾

Recommended Action

MANUAL REVIEW × BLOCK × × ▾

Transaction Risk Checker

Enter an existing transaction ID to investigate its ML risk, graph exposure, network propagation, and explanation.

Transaction ID

T027746 ▾

Analyze Transaction

Transaction T027746

Unified Risk

16.11/100

ML Fraud Probability

3.24%

Graph Risk

41.67%

Network Risk

25.00%

Risk Category:

Low

Recommended Action:

APPROVE

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Investigation Filters

Risk Category

High ×

Critical ×

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Recommended Action

MANUAL REVIEW ×

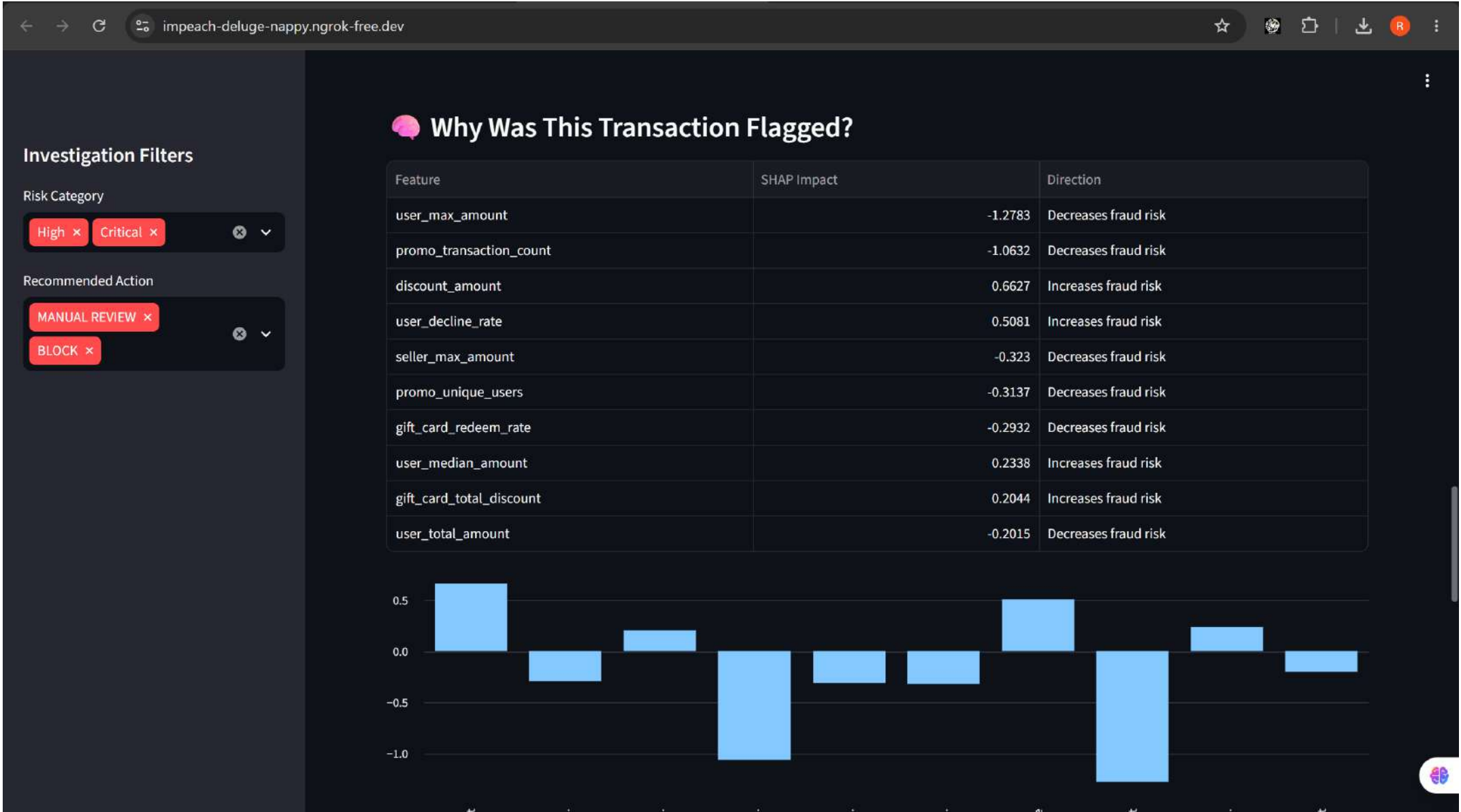
BLOCK ×

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Transaction Details

Field	Value
transaction_id	T027746
user_id	U03915
seller_id	S0167
device_id	D02616
ip	122.127.57.16
amount	1223.39
timestamp	2026-03-08 22:54:19.717582
shipping_address	26788 Stacy Land Apt. 473, Adrianaborough, TX 31935
billing_address	26788 Stacy Land Apt. 473, Adrianaborough, TX 31935
card_token	CARD_01818



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Investigation Filters

Risk Category

High ×

Critical ×

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Recommended Action

MANUAL REVIEW ×

BLOCK ×

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High-Risk Users

	user_id	transactions	max_risk	avg_risk	fraud_cases
0	U00966	8	95.1161	77.5069	4
1	U00073	2	94.555	94.0706	2
2	U00231	1	93.3338	93.3338	1
3	U03971	6	93.2659	84.541	5
4	U03962	1	92.6845	92.6845	1
5	U01021	4	92.5678	91.2552	4
6	U03061	8	92.3029	73.2938	5
7	U02150	1	92.2361	92.2361	1
8	U02929	3	92.21	86.2692	3
9	U03807	5	91.5736	90.6611	5

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Investigation Filters

Risk Category

High ×

Critical ×



Recommended Action

MANUAL REVIEW ×



BLOCK ×



Detection Architecture

Abuse Sentinel Pipeline

Transaction Data ↓

Behavioral Feature Engineering ↓

XGBoost Fraud Model ↓

Graph Construction ↓

Graph Risk Analysis ↓

BFS Risk Propagation ↓

Unified Risk Score ↓

SHAP Explainability ↓

Approve / Monitor / Manual Review / Block

The system combines transaction-level machine learning with network-level fraud detection to identify coordinated fraudulent behavior.



Abuse Sentinel fraud detection pipeline is operational.

